Junyeob Baek

Ph.D. Student @ KAIST

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Research Interest

My research goal is to build *"An embodied machine that can develop intelligence throughout its lifetime, just like humans."* Specifically, the machine should establish and grow its own lens of worlds, goals, and values in open-ended environments — with interpretability and alignment as guiding concerns.

My current path toward this vision is to build **compositional world models** — understanding the world as reusable structure — and the **recursive reasoning** that runs over them to generalize beyond experience.

I pursue the following research questions, with **Q1** as my current focus throughout the PhD:

- **Q1. How can AI understand the world more systematically and compositionally? – Systematic World Understanding & Generalization (current focus)**
- *Q2. How can AI continuously accumulate and apply knowledge/skills? – Continual Learning*
- *Q3. Can AI develop its own goals following intrinsic motivation? – Autotelic Agents*
- *Q4. How can AI maintain autonomy while sharing human values? – Human-Value Alignment*

Education

Mar. 2025 – Present — **Korea Advanced Institute of Science and Technology(KAIST)**
Ph.D. in Computer Science - **Advisor:** Sungjin Ahn — Daejeon, Korea

Mar. 2023 – Feb. 2025 — **Korea Advanced Institute of Science and Technology(KAIST)**
M.S. in Artificial Intelligence - **Advisor:** Sungjin Ahn — Daejeon, Korea
- **Thesis**: Dreamweaver: Learning Compositional World Models from Pixels

Mar. 2014 – Feb. 2018 — **Hanyang University(ERICA)**
B.S. in Robot Engineering - **GPA** - **4.29 / 4.5** — Ansan, Korea
- *Summa Cum Laude*
- *Capstone Project1: Disaster and Rescue Robot Simulation Modeling(Darrsm) - 2nd Prize in College*
- *Capstone Project2: Walk Yourself, Toddler!(WalkYTo) - 3rd Prize in Department*

Research Experience

Jun. 2025 – Aug. 2025 — **Visiting Researcher**, Mila-Quebec AI Institute — Montreal, Canada
- Collaborating with *Yoshua Bengio*'s research group and *LawZero* team members
- Conducting research project on **Mechanistic Interpretability and Chain-of-Thought Faithfulness in LLMs**
- **Main Collaborators:** *Joumana Ghosn*, *Pierre-Luc St-Charles*, *Marc-Antoine Rondeau*, *Minsu Kim*

Honors & Awards

2026 **Best Paper Award**, ICML 2026 Workshop on Compositional Learning

2026 **Oral Presentation (Top 0.7%, 168/23918)**, ICML 2026

2026 **Silver Reviewer**, ICML 2026

2025 **Outstanding Teaching Assistant Award**, KAIST

Publications

*Equal contribution

Conference Papers

C5 **Junyeob Baek***, Mingyu Jo*, Minsu Kim, Mengye Ren, Yoshua Bengio, and Sungjin Ahn. "Generative Recursive Reasoning". In: *Conference on Neural Information Processing Systems (NeurIPS)* (2026).

C4 Doojin Baek*, **Junyeob Baek***, Mingyu Jo, Hosung Lee, Taegu Kang, Gyubin Lee, and Sungjin Ahn. "BabyTheorist: A Benchmark for Learning to Theorize the World from Observation Alone". In: *Conference on Neural Information Processing Systems (NeurIPS)*, Evaluations & Datasets Track (2026).

C3 Doojin Baek*, Gyubin Lee*, **Junyeob Baek**, Hosung Lee, and Sungjin Ahn. "Learning to Theorize the World from Observation". In: *International Conference on Machine Learning (ICML)* (2026). **Oral Presentation (Top 0.7%)** • **Best Paper Award**, ICML 2026 Workshop on Compositional Learning.

C2 **Junyeob Baek**, Yi-Fu Wu, Gautam Singh, and Sungjin Ahn. "Dreamweaver: Learning Compositional World Models from Pixels". In: *International Conference on Learning Representation (ICLR)* (2025).

C1 Chang Chen, **Junyeob Baek**, Fei Deng, Kenji Kawaguchi, Caglar Gulcehre, and Sungjin Ahn. "PlanDQ: Hierarchical Plan Orchestration via D-Conductor and Q-Performer". In: *International Conference on Machine Learning (ICML)* (2024).

Workshop Papers

W1 **Junyeob Baek**, Hosung Lee, Christopher Hoang, Mengye Ren, and Sungjin Ahn. "Discrete JEPA: Learning Discrete Token Representations without Reconstruction". In: *International Conference on Machine Learning (ICML) Tokenization Workshop* (2025).

Preprints

P1 Donghwan Chi, Hyomin Kim, Yoojin Oh, Yongjin Kim, Donghoon Lee, Daejin Jo, Jongmin Kim, **Junyeob Baek**, Sungjin Ahn, and Sungwoong Kim. "Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM". *arXiv preprint arXiv:2505.17726* (2025).

Thesis

Junyeob Baek. "Dreamweaver: Learning Compositional World Models from Pixels". Master's Thesis, KAIST, Advisor: Sungjin Ahn (Feb. 2025).

Academic Service

Conference/Journal Reviewer
International Conference on Learning Representations (ICLR) 2025–2027
International Conference on Machine Learning (ICML) 2026 — **Silver Reviewer**
Conference on Neural Information Processing Systems (NeurIPS) 2026
Transactions on Machine Learning Research (TMLR) 2026–

Workshop Organizer
3rd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), Mila, Aug. 2026, Link
2nd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), Mila, Jul. 2025, Link

Teaching Experience

Spring 2026 **Teaching Assistant**, CS30707: Introduction to Reinforcement Learning, KAIST

Fall 2025 **Teaching Assistant**, CS672: Reinforcement Learning, KAIST

Spring 2025 **Teaching Assistant**, CS377: Introduction to Reinforcement Learning, KAIST
- Outstanding Teaching Assistant Award recipient

Professional Experience

Jun. 2021 – Jun. 2022 **MakinaRocks**, ML Research Engineer(Robotics Focused) Seoul, Korea
- **Optimization of Welding Task and Motion Planning in Vehicle Manufacturing Process**
  - Challenges: 1. Large-scale and high complexity in Optimization Problem. 2. Required high-accuracy planning in highly cluttered environments. 3. Required tight Time Window for control
  - [*Contribution 1*] GA-based Multi-objective Optimization Module for Welding-point Assignment
  - [*Contribution 2*] Multi-Robot Interlock Scheduling System with Swept-volume-based collision detection

Dec. 2017 – Jun. 2021 **Syscon, Co., Ltd.**, Robotics Software Engineer/Research Strategy Manager Incheon, Korea
- **AMR Navigation System Design & Product Development**
- [*Field Exp.*] **Successful Deployment for Smart Factory Project**
  - More than 300 delivery tasks per day with 14 robots
- [*Research Exp.*] **Learning-based High-accuracy Object Pose Estimator(Dock Pose Estimation)**
- [*Research Exp.*] **Distilled Meta Policy Navigation System for Multi-behavior Skillset and its Strategies**